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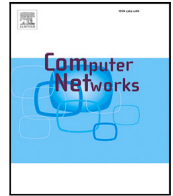
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Review article

The Evolution of Mobile Network Operations: A Comprehensive Analysis of Open RAN Adoption

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ABSTRACT

As Mobile Network Operators (MNOs) seek to revolutionise their infrastructures and meet the growing demands of a data-centric world, the adoption of open RAN (Radio Access Network) technology has emerged as a compelling solution. This paper offers a comprehensive exploration of open RAN's transformative potential within the MNO landscape and investigates the concrete implementation opportunity of Open-RAN (O-RAN). Starting out by providing an exhaustive overview of the state of the art in open RAN technology, research and deployments, leads to an in-depth analysis of the decision roadmap of open RAN adoption. This roadmap considers the network design, vendor criteria and implementation strategy. This paper contributes with a comprehensive examination of components and an overview of increasingly complex design opportunities including functional split and accelerator options. Thus, it serves as an enlightening and pragmatic guide for MNOs navigating the transition to opening the interfaces in their RAN. Finally the paper proposes a pragmatic exploration of implementation strategies, providing insights to the drawbacks MNOs will face and proposed solutions. Hence, this paper highlights how the O-RAN system is built in modules, which adds additional importance to the increasingly complex design phase when multiple aspects must be evaluated to become the (r)evolution of mobile networks.

1. Introduction

Mobile networks are critical infrastructure used for countless purposes including Internet of Things (IoT), telemedicine and high-resolution video streaming. The global mobile network data traffic has already exceeded 120 EB per month, and is expected to triple within the next five years [1]. This continuously increasing data traffic requires Mobile Network Operators (MNOs) to endlessly upgrade and expand their networks. Network expansions and upgrades are expensive, since they involve installation of new sites in the network, or refinement of existing ones. Hence, MNOs are looking for other opportunities to decrease the cost of running a mobile network and currently, the mobile industry is booming with promises of what open Radio Access Networks (RAN) can offer the market.

The traditional way of building the RAN part of a mobile network is to install an antenna, Radio Unit (RU) and Baseband Unit (BBU) on the selected site position. This is referred to as a distributed RAN installation. The antenna and the RU are connected by a coax cable, and located somewhere high, for instance in the top of a mast. The RU is then connected to a BBU via a fibre connection. The RAN installation is shown in Fig. 1, together with its connecting interfaces. These interfaces being the fronthaul connecting the RU to the BBU and the X2/Xn

interface connecting the BBUs on different sites. The RU interacts with the BBU over the Common Public Radio Interface (CPRI) protocol. The CPRI protocol is openly available for various vendors to contribute to the ecosystem, however; it does not limit product differentiation on the management side [2]. This leaves the end products communicating over the CPRI protocol to be proprietary installations. Hence, this makes a RU from one vendor limited to interact with a BBU from the same vendor or from a vendor knowing the specifics of the first vendor's management plane. Given the vision of a more diverse vendor ecosystem; the open RAN ambition is publicly open interfaces in the RAN. This enables components from a limitless number of vendors to inter-operate on a per site level. Hence, MNOs not only seek diversity in equipment vendors for neighbouring sites but also to have equipment from different vendors on the same site.

This paper draw a sharp distinction between, open RAN as the concept of open interfaces in the RAN and Open-RAN (O-RAN) which is a concrete implementation option of open RAN. The industrial alliance, O-RAN Alliance creates specifications based on the vision of open RAN interfaces and a multi-vendor environment. The specifications created by O-RAN embed the three concepts; open RAN, Virtual RAN (vRAN)

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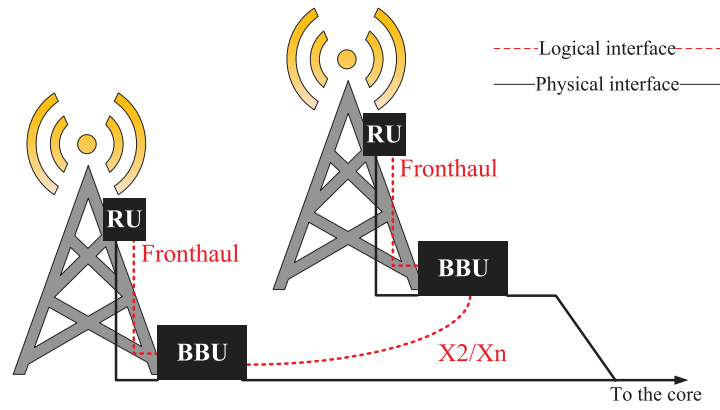


Fig. 1. The distributed RAN installation, having both Radio Unit (RU) and Baseband Unit (BBU) installed on the cell side. The figure illustrates the logical and physical communication between RU and BBU on-site and between BBUs on different sites, which must be open to achieve open RAN. The figure has a third dimension, proposing scalability by the use of Cloud-RAN (C-RAN) architecture.

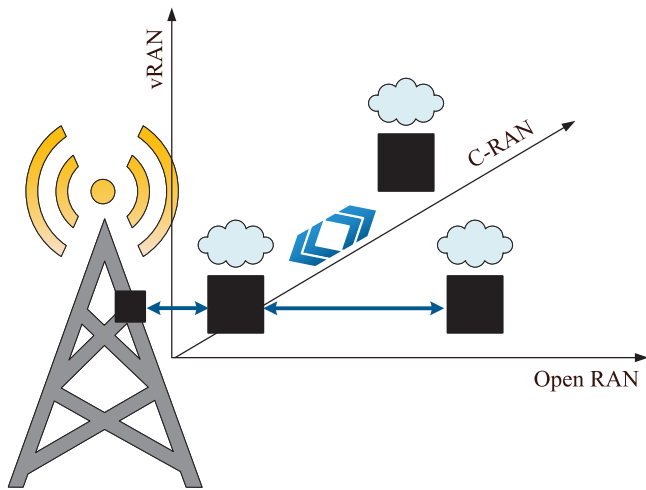


Fig. 2. The concepts of open Radio Access Networks (RAN) and virtual RAN (vRAN) together with Centralised RAN (C-RAN). vRAN providing vertical division of components into functionalities and hardware and open RAN leading to a horizontal division of different components. In vRAN the hardware (HW) and software in the form of Distributed Unit (DU) and Centralised Unit (CU) functions are connected using an abstraction layer (ABS). C-RAN adds the third dimension of scalability in the network.

and Cloud-RAN (C-RAN) providing the option of centralised processing. Additionally, they follow a trend where the baseband processing functions are divided into two components; the Distributed Unit (DU) and the Centralised Unit (CU). This way, the DU handles part of the physical layer and the more time-critical functions in the lower part of the radio protocol stack. Whereas, the CU handles control functions and interoperability with other sites. Furthermore, the CU functions can be divided into a user and a control part. The exact division of functions between the DU and the CU is referred to as the functional split, which is a concept that becomes more vital when centralising some of the baseband functions.

The concept of vRAN separates the software and the hardware of the individual RAN components. Hence, instead of a BBU being one component, it becomes a set of software functions that can run on hardware from another vendor. To visualise the difference between vRAN and open RAN; vRAN is defined as the vertical openness of the RAN components, where open RAN is the horizontal openness of the RAN components. This is illustrated in Fig. 2. A further abstraction of this definition is to add a third dimension, where a centralisation of the processing, as known from C-RAN, will provide scalability to the network, as shown in Fig. 2.

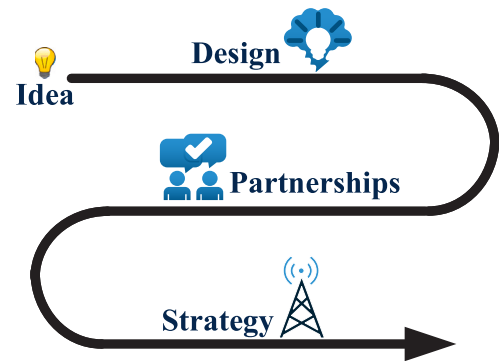


Fig. 3. The MNO decision roadmap illustrating the progression of this paper.

This paper presents the MNO decision roadmap towards open RAN implementation, as well as the design complexities a MNO will face. The structure of the paper is illustrated in Fig. 3, emphasising how this paper first presents the design criteria, then criteria for partnerships and finally criteria for selecting the right cells. Hence, it provides a set of guidelines towards open RAN decision making as well as outlines pros and cons of open RAN adoption. Thus, in the remainder of this paper, Section 2 provides an overview of the state of the art in open RAN concept, research, standardisation and deployments. In Section 3, the concept of open RAN and various elements of the O-RAN ecosystem are explored. Section 4 presents expectations from a MNO point of view as well as selection criteria for choosing vendor partnerships. Section 5 provides an implementation strategy which is examined in a case study. Section 6 discusses the potentials and pitfalls of open RAN and finally the conclusion closes this paper.

2. State of the art

Much work is currently going on in the process of opening up RAN interfaces. Thus, this section presents the state of the art in terms of open RAN interfaces within the areas of research, standardisation, testing and deployments. Furthermore, we also present an overview of industrial alliances and highlight our own contribution to the community.

2.1. Related work

Work in the field of open RAN can be divided into five categories: Surveys, deployment oriented, testbeds, network intelligence and MNO perspectives.

The concept of open RAN has been considered in several *surveys*, namely in [3,4], very comprehensive studies are provided. These studies give insights to the various working groups of O-RAN Alliance and their individual contributions, as well as giving a detailed overview of the O-RAN ecosystem. Furthermore, the work in [5], besides showing an automated way of xapp deployment, also provides a good overview of standardisation efforts and current work in the field of O-RAN. The work in [6] also provide a good overview of the ecosystem, along with proposing the use of zero-forcing beamforming. The concept of open RAN is further surveyed in [7,8], and [9] focus on open RAN for 6G.

Within *deployment oriented* research, the work in [10] presents an overview of O-RAN, vRAN and C-RAN as well as results from a vRAN realisation. Results show a high resource utilisation improving the scalability of the RAN [10]. Authors in [11] presents a tool that can navigate in vRAN and open RAN to realise efficient deployment. The work in [12] utilises the intelligence of the O-RAN system to improve cell selection based on channel quality and bitrate. Authors in [13] study the placement of virtualised DUs by taking factors like cost and delay into account. The work in [14] highlights challenges faced in xapp development.

In the field of more practical *testbed* research contributions, the work in [15] addresses the complications related to designing hardware ready to cope with the requirements of 5G. Authors in [16] present a comprehensive open RAN experimentation platform. The work in [17] expands an existing test platform to enable Machine Learning (ML) research at scale using O-RAN components and programmable base stations. In [18] authors utilise the NS3 simulation tool for an O-RAN testbed. Also the work in [19] utilises a testbed to verify their proposed solution for edge core deployment.

The area of designing *intelligent networks* receive much attention. The work in [20] analyses three RAN Intelligent Controller (RIC) designs based on key performance metrics. Authors in [21] propose an algorithm to prevent conflicts happening when various applications contribute to the control of the network. Furthermore, multiple works focus on adding Artificial Intelligence (AI) into open RANs [22–27].

Papers in the area of *MNO perspectives* are yet very limited, however the work in [28] elaborates on the implications of installing an end to end O-RAN network including the cloud native infrastructure and the transport network. The work in [29] reports on several lessons learned from open RAN deployment including: new engineer skill-sets needed and the need for system level Service Level Agreements (SLA).

2.2. Research projects

Open RAN is a very timely topic, and thus, many on-going research activities are directed towards creating open interfaces for mobile networks. An overview of some of these projects can be found in Fig. 4, where the projects are divided into three main categories: testbeds/trials, hardware/R&D activities and O-RAN specific contributions. These projects are briefly described below.

2.2.1. Testbeds/trials

- The “Building Reusable testbed Infrastructures for validating Cloud-to-device breakthrough technologies”, 6G-BRICKS project, aims to establish an experimental research facility to evaluate key 6G candidate technologies. In the testcenter, initial integrations with O-RAN are performed, aiming for the future-proofing and interoperability of outcomes [30].
- The “Field Trials beyond 5G”, FIDAL project, aims at creating the infrastructure for the future of 5G and 6G by promoting open architectures, large experimentation sites and taking a multi-stakeholder approach [31].
- The “reEaLizing healthcare 4.0 eXploiting the 6G network evolution”, ELIXIRION project, will create the first fully-integrated, inter- and multi-disciplinary, highly-innovative training and research network that will set the foundations of the emerging

Healthcare 4.0 paradigm by leveraging 6G technologies. This includes creating a sustainable open market, easing access for new players [32].

- The “6G eXperimental Research infrastructure to enable next-generation XR services”, 6G-XR project, will develop an evolvable experimental infrastructure that demonstrates the performance of key Beyond 5G (B5G)/6G candidate technologies, components, and architectures to keep the infrastructures valid now and in the longer run. It will also validate a representative end-to-end B5G/6G architecture including end-to-end service provisioning with slicing capabilities, and at cloud implementation level including open RAN [33].

2.2.2. Hardware/R&D activities

- The “European Core Technologies for Next Generation Communication-Computing Hardware”, COREnext project, aims to develop core technologies for Beyond 5G and 6G adoption that supports an open, multi-vendor and multi-tenant disaggregated RAN by employing virtualisation technology [34].
- The “Realising Enabling Architectures and Solutions for Open Networks”, REASON project, brings together an ecosystem representing the entire telecommunication R&D supply chain, including three major mobile network equipment vendors to develop a roadmap for open 6G networks, which will set the framework for new developments across the entire technology stack [35].
- The “Towards Ubiquitous 3D Open Resilient Network”, TUDOR project, will research and develop open network components and their seamless interoperability in the wider RAN, core, transport network environment and service platforms [36].
- The “Yorkshire Open RAN”, YO-RAN project, brings together two leading Yorkshire universities with a number of suppliers and network operators, several based in and around Yorkshire, to develop Open RAN components and a RIC for Neutral Host Networks [37]. It has not been possible to find an end-date for the YO-RAN project, thus; on Fig. 4 it is fading out on the right hand side.
- The “Distributed Artificial Intelligence-driven Open and Programmable Architecture for 6G Networks”, ADROIT6G project, aims to build the future architecture of mobile communications to serve emerging stringent requirements of innovative extreme future-looking applications [38].
- The “Empower Converged Optical Wireless Configurations with Cell-Free Technologies for High-Density 6G Networks”, EMPOWER 6G project, proposes an architecture that is in-line with O-RAN Alliance for design of cell-free network mechanisms [39].

2.2.3. O-RAN specific contributions

- The O-RANOS project works towards addressing key architectural and technological challenges for deploying end-to-end O-RAN multi-domain (private and public) interoperable network solutions [40].
- The “Beyond 5G Energy Efficient Networking by Hardware Acceleration and AI-Driven Management of Network Functions”, BEGREEN project, investigates the opportunity for adding an AI-assisted energy-aware “Intelligent Plane” as an additional plane, that allows the data, model, and inference to be seamlessly exchanged between network functions to extend the O-RAN architecture to address energy optimisation [41].

2.3. Industrial alliances

In an ecosystem of open interfaces, it is beneficial to collaborate in many ways. Thus, multiple industrial alliances have aroused with the purpose of create open interfaces in RANs. Some of these are:

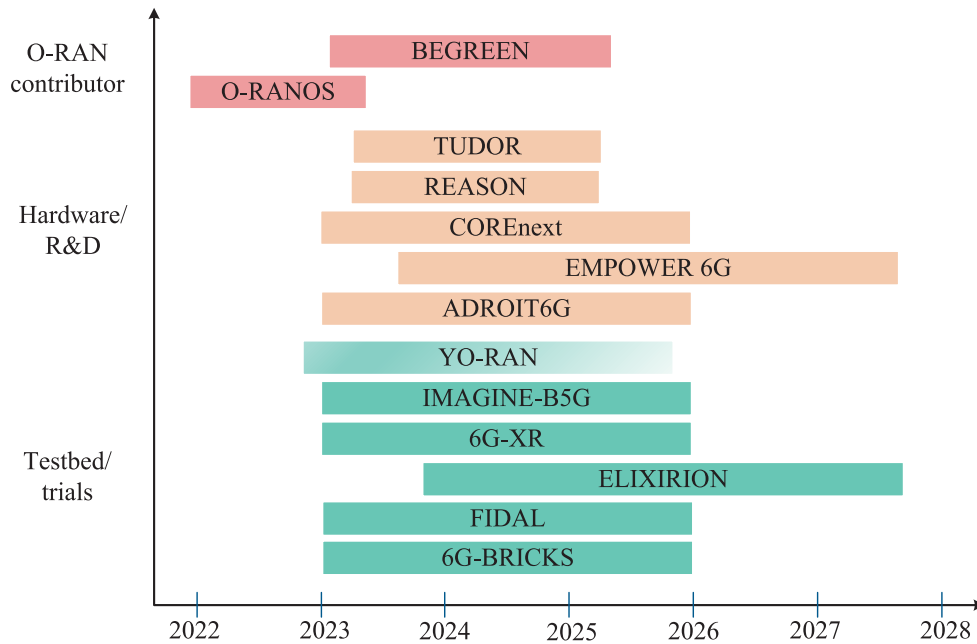


Fig. 4. Various projects supporting the development of open RAN, divided into three groups: testbeds/trials, hardware/R&D activities and O-RAN specific contributions.

- O-RAN Alliance is a world-wide community of mobile operators, vendors, and research and academic institutions with the mission to re-shape Radio Access Networks to be more intelligent, open, virtualised and fully interoperable. The alliance counts 33 MNOs and more than 300 members in total [42].
- OpenRAN is a project in TIP, that works towards accelerating innovation and commercialisation in RAN domain with multi-vendor interoperable products and solutions easy to integrate in the MNO's network and verified for different deployment scenarios [43].
- Open RAN Industry Alliance (ORIA), is a national body in South Korea with a strategic plan to develop a domestic open RAN ecosystem in the country. The alliance counts three MNOs and 30 members in total [44].
- Open RAN Policy Coalition has the mission to promote policies that will advance the adoption of open and interoperable solutions in the RAN by creating innovation, spur competition and expand the supply chain for advanced wireless technologies including 5G. The coalition consists of more than 50 members [45].

2.4. Open RAN testing

Globally there are multiple options for testing open RAN and O-RAN solutions and products.

- O-RAN ALLIANCE defines an overall approach to testing and integration of O-RAN products and solutions, including criteria and guidelines for Open Testing and Integration Centre (OTIC). Currently 17 OTICs exist globally [46].
- Colosseum simulator is the world's largest wireless network emulator with hardware in the loop [47]. Colosseum empowers researchers with the tools to capture and reproduce different conditions of wireless channels, and to experiment at scale [4].
- OpenRAN Gym is a publicly available research platform for data-driven O-RAN experimentation, based on the Colosseum as experimental platform [48].
- C-DOT O-RAN Test and Integration Lab provides around 1000 m² indoor and 500 m² outdoor open RAN testing facilities in India [49].

- Arena is an indoor testbed, connected to 64 antennas deployed across a 200m² office space, which provides ideal conditions for edge computing and private network testing [4].
- The US National Science Foundation PAWR program provides a number of city-scale platforms for open RAN testing. These consist of [4]:
 - POWDER: for sub-6 GHz cellular deployments [50]
 - COSMOS: for mmWave communications [51]
 - AERPAPW: for aerial cellular deployments [52]

2.5. Standardisation efforts

The O-RAN alliance is creating specifications, meaning that they provide a recommendation of how they find networks can be realised. A standard is when a number of groups join forces and agree that this is the right way to do things. For instance 3GPP consists of seven partners that needs to transpose a technical specification to become a standard. 3GPP has already standardised several interfaces in the network, including E1 connecting the CU-User Plane (UP) and the CU-Control Plane (CP) and F1 connecting the DU to the CU as well as the Uu interface between the UE and the RAN [3]. Furthermore, which is important from an open RAN point of view, 3GPP has also standardised the X2/Xn interface, connecting the CU's on different sites. The reason why this interface is important to consider, is because it enables the MNO to have CUs from different vendors on neighbouring sites. However, for this to work well, a management system that supports multiple vendors is required. Status with regard to standardisation of the O-RAN specifications, is that the European 3GPP organisational partner, ETSI has adopted the O-RAN specification for fronthaul control, user and synchronisation plane [53]. Furthermore, the O-RAN alliance has signed Memorandum of Understandings (MoUs) with 3GPP organisational partners ATIS [54] and TSDSI [55].

2.6. Commercial trials and deployments

The first commercial deployments of per-site multi-vendor networks have been launched within the last four years, where Rakuten in Japan, 1&1 in Germany and Dish in the US are the ones most discussed in the medias. Those deployments are not based on any public available

specifications. 1&1 are using the Rakuten platform for end-to-end integration of the network which is Europe's first fully virtualised mobile network using products from more than 80 providers [56]. On the other hand, commercial deployments based on the O-RAN specifications are also being rolled out today, for instance by Vodafone in five countries, Telefonica in three countries and Deutsche Telecom in Germany [57]. NTT DoCoMo in Japan has also adopted the O-RAN fronthaul specifications and by March 2022 they had deployed 30,000 O-RAN sites [58]. Many more deployments and trials are ongoing, an overview can be found in [58]. In downtown New York a trial has been carried out for OrchestRAN; a novel orchestration framework for next generation systems that embraces and builds upon open RAN [47]. In 2022 India's first O-RAN based live 5G network validation was conducted achieving speeds over 1 Gbps [59]. Another notable trial is the open RAN PoC in South Africa, Nigeria and Liberia also based on the Rakuten platform. Latest news in the field of open RAN, is that AT&T have announced they will partner up with Ericsson aiming to move 70% of the MNOs wireless network traffic to open-capable platforms by late 2025 [56].

2.7. Our contribution

This work provides a MNO overview and guide to open RAN. It does not provide a thorough description of the O-RAN ecosystem, as this work refer to other great papers for that matter (see subsection 2.1). The paper provides insights in the most recent technical discussions on how complexity of O-RAN evolves and elaborates on what the various opportunities will bring. Our contributions can be divided into several key areas, all of which significantly enhance the understanding of the subject matter.

- A thorough State of the Art section including an overview of research projects and industrial alliances. We delve into the evolving open RAN landscape by incorporating recent research projects, testing, deployment and industrial alliances. By doing so, we offer a comprehensive view of the latest developments, innovations, and trends that are currently shaping the field.
- An exhaustive understanding of open RAN Components and network design opportunities on top of that the increasing complexity. Section 3 is dedicated to dissecting the various components related to the open RAN architecture. We examine how open RAN and O-RAN differs from traditional RAN systems and emphasise the various design options MNOs can face. This includes an extensive overview of evolution RUs and hardware for vRAN.
- MNO Selection Criteria, to be used for inspiration to MNOs selecting vendors and equipment. This paper delves into the critical decision-making process that MNOs face when selecting specific components or vendors within the O-RAN ecosystem. We outline a set of criteria that MNOs can consider, thus; this section provides valuable insights into the factors that influence MNOs' choices in adopting the concept of open RAN.

3. Network design

The introduction of open interfaces in the RAN will bring new ways of thinking for the MNOs. This is particularly within the areas listed below which will be elaborated within the following subsections. This includes an alignment of which design options are necessary for the MNO to consider when adopting a new RAN architecture. The first part is more generic related to the conceptual term of open RAN:

- Degree of openness
- Network management
- Open interfaces

The Latter part is merely related to the implementation of the O-RAN ecosystem, beginning with a brief description of the system. Then it touches down on the following themes:

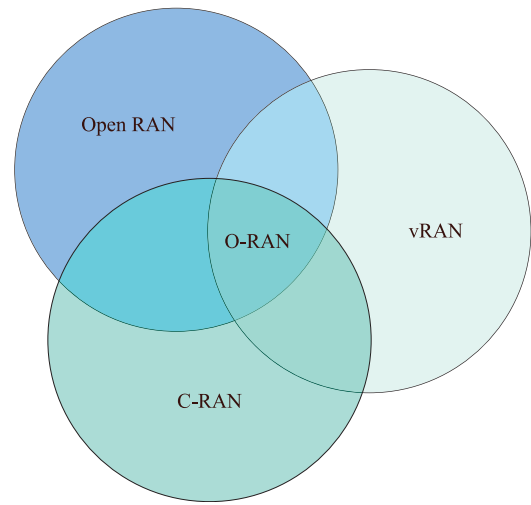


Fig. 5. An abstraction of the O-RAN ecosystem showing how it embeds the concepts of Open RAN, Cloud-RAN (C-RAN) and virtual RAN (vRAN).

- Virtualisation
- Centralisation
- Open Fronthaul
- Native AI
- Intelligence
- Security

The difference between the conceptual term of open RAN and the O-RAN ecosystem implemented in specifications is illustrated in Fig. 5.

3.1. The open RAN concept

The concept of open RAN defines open interfaces in the RAN, enabling RAN components from different vendors, on the same site to inter-operate. This vision is to create new opportunities for the MNOs to mix and match equipment on a per site level. Hence, in practice the open RAN concept does not define anything about the components of the RAN, they can be proprietary, distributed or centralised as long as the interface between RU and DU+CU is publicly available, for an indefinite number of vendors to be able to contribute.

3.1.1. Degree of openness

Different MNOs can have various expectations to a future open RAN installation, is vRAN for instance an open RAN network if different vendors are used for hardware and software? Hence, openness in terms of network interfaces can be understood to many extends, and this paper uses the following terms:

- Single vendor open RAN: a single vendor delivers a network based on open interfaces. This way the MNO only has one vendor to manage, and only one vendor is responsible for the integration. However, the MNO has the leverage of being able to install equipment from other vendors at any time.
- vRAN: In vRAN the same vendor is delivering the software to all RUs, DUs and CUs, and this software can run via an abstraction layer on 3rd party Commercial off the Shelf (COTS) hardware. The fronthaul interface can be open or not, and the number of vendors can be one or multiple.
- Multi-vendor RAN: In this case, multiple vendors contribute to the network, but only the contributors knows the specifics of the interface, since the information is not publicly available. Hence, no other vendors can contribute to the network.

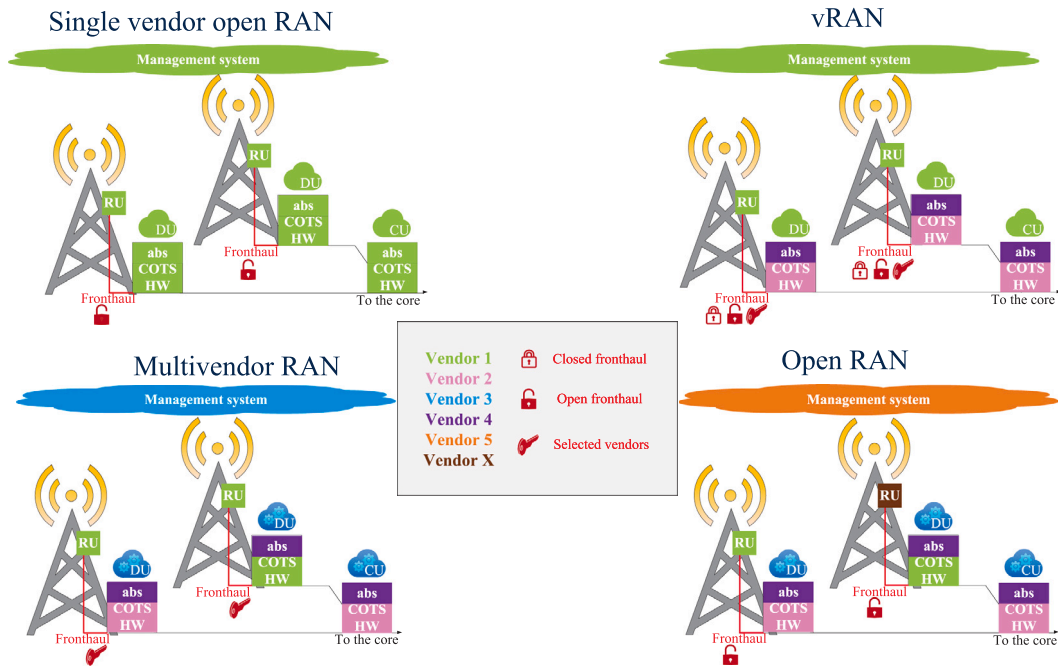


Fig. 6. Deployment options for open Radio Access Network (RAN), where all options utilise virtualised network functions, requiring an abstraction layer (ABS) between the hardware (HW) and Distributed Unit (DU) or Centralised Unit (CU) functions. The closed lock refers to a proprietary fronthaul, the open lock refers to open fronthaul and the key refers to selected vendors being able to contribute to the ecosystem.

- Open RAN: The network has open interfaces, based on publicly available information. Hence, there is no technical limitations for vendor selection.

The various levels of openness listed are distinguished based on the openness of the fronthaul network. If the fronthaul network is open, it is a mix and match situation where any vendor can contribute to the ecosystem. If the fronthaul is closed it is a proprietary solution where only one vendor delivers RU and DU/CU. There is also a solution in-between where only a selected group of vendors have information about the fronthaul interface, and in this case, only the selected group of vendors can contribute to the ecosystem. The various levels are illustrated in Fig. 6. As the figure shows, the closed fronthaul is illustrated by a closed lock, the open fronthaul is illustrated by an open lock and the selected vendors option is illustrated by a key. Fig. 6 shows how vRAN can support a proprietary solution, selected vendors solution and open fronthaul. The single vendor open RAN solution would be the same as vRAN with open fronthaul, but having only one vendor. The multi-vendor RAN, can be multi-vendor with selected vendors or it can be open RAN with an open fronthaul. The figure illustrates only the conceptual architectures, and not specific implementations like the O-RAN ecosystem.

3.1.2. Network management

Introducing open interfaces in the RAN will leave a whole new role for the MNO to fill out: Network management of components from a variety of different vendors. Thus, it requires the MNO to not only agree on the selection criteria but also to possess skills within integration of network components. Another aspect to meet in the selection criteria, is the company selected as vendor, do they as a company live up to our expectations for instance regarding Corporate Social Responsibility (CSR) or sustainability? Hence, many new workflows are to be implemented when the MNO not only becomes his own network manager, but also of equipment from multiple vendors. This opens up opportunities for new business areas and calls for automated processes. Processes that can handle operations management including fault, configuration, performance, and security management of open RAN networks [60].

3.1.3. Open interfaces

The open RAN concept is defined by open interfaces in the RAN. The interface between RU and DU in particular, which today is using the CPRI protocol, and is only partly open. In order to reduce the requirements to the fronthaul interface connecting the RU to the DU, more functions can be added to the RU. This is referred to as the Functional Split (FS), and even though the FS between DU and CU is already defined [61], then the low layer split separating the RU and the DU has not been standardised yet. In terms of open interfaces, the FS is crucial, because it determines which type of data is carried on the fronthaul network, if it is raw I/Q symbols as in the case of the CPRI protocol, subcarriers or codewords [62]. Fig. 7 compares the FS options from various alliances; 3GPP, CPRI, O-RAN and Small Cell Forum (SCF). Considering the FSs proposed by O-RAN alliance 7-2x cat A and B, then some of the other proposed FSs have more functions included in the RU, leading to more complex RUs but also lowered requirements on the fronthaul interface. Furthermore, a specific functional block is important to consider, the resource element mapper, which detects unused subcarriers and makes the load on the fronthaul network vary by the user load [62]. Hence, FSs having the resource element mapper in the DU, including O-RAN FS 7-2x, will provide a fixed occupation on the fronthaul interface. Whereas, FSs having the resource element mapper included in the RU will only occupy the fronthaul interface if actual traffic is sent. However, it must be noted, that all FSs have benefits and complications. Hence, there is a clear trade-off to be considered between RU complexity and requirements to the fronthaul transport network.

3.2. The O-RAN ecosystem

The O-RAN specifications define a specific implementation of a RAN build on the vision of openness and intelligence [42]. The O-RAN ecosystem is a new way of utilising open interfaces where the system is designed having in mind the openness, intelligence and vision for future mobile communication. This has resulted in a system embracing not only the open RAN concept, but also virtualised functions in the RAN and the opportunity for centralised processing, as illustrated in Fig. 5. The O-RAN specifications include several elements [63]:

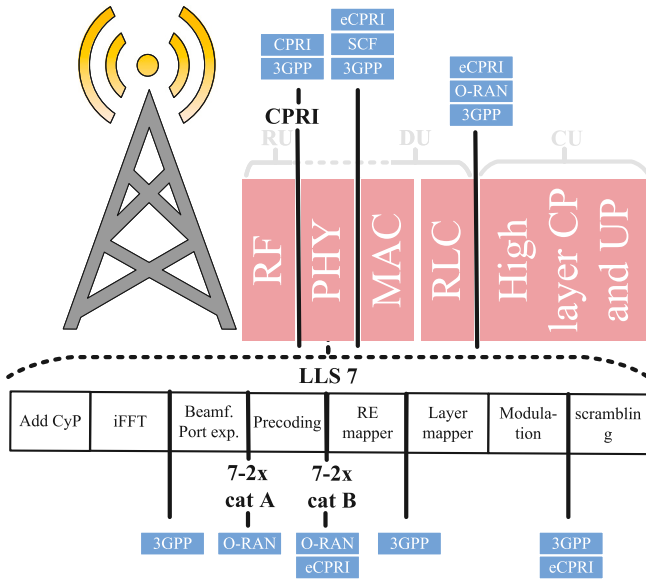


Fig. 7. The Functional Split (FS) between Radio Unit (RU) and Distributed Unit (DU), as proposed by various standardisation organisations and alliances, including the original FS used for the Common Public Radio Interface (CPRI) and FS 7-2x proposed in O-RAN specifications.

- Re-thinked proposals for RU, DU and CU, now referred to as open RU (O-RU), open DU (O-DU) and open CU (O-CU)
- Network elements required for virtualisation, being the Service Management and Orchestration Framework (SMO)
- New network elements for intelligence: The RICs, which are used differently for near real time (RT) and non RT

Fig. 8 presents a high level drawing of the O-RAN architecture which embeds the concept of RAN virtualisation and includes RICs handling near RT functions and AI processing. The logical presentation of the O-RAN ecosystem includes multiple new interfaces for intra-connecting the RAN, this is illustrated in Fig. 9. The various interfaces are described below [42]:

- The Open Fronthaul (O-FH) connects the O-RU to the O-DU and the SMO framework.
- The F1 interface connects the O-RU to the O-CU, both to CP and UP.
- The E2 interface connects the O-DU and O-CU to the near RT RIC.
- The A1 interface connects the near RT RIC to the SMO framework.
- The O1 interface connects the SMO to the O-DU, O-CU and the near RT RIC.
- The O2 interface connects the hardware, referred to as the O-cloud to the SMO. Authors in [60] developed a network management system that connects the O-Cloud and RAN functional elements to the SMO, via the O-RAN defined O1 and O2 interfaces.

Apart from the interfaces mentioned above, the system also includes the X2/Xn interface for communication between CUs [64], the E1 interface connecting the O-CU CP to the O-CU UP [42] as well as the NG interface connecting to the 5G core [65], however; these are maintained by 3GPP. In Table 1, all interfaces are divided after those who maintain them, being O-RAN alliance and 3GPP.

3.2.1. Virtualisation

Virtual network functions, in the form of vRAN, plays a large role in the foundation of O-RAN specifications. Furthermore, they open up for many new vendors to enter the market, since new functions and tasks

Table 1

Interfaces in the O-RAN ecosystem, divided into those maintained by O-RAN Alliance and those standardised by 3GPP [64].

Maintained by O-RAN:	Maintained by 3GPP:
A1	E1
O1	F1
O2	NG
E2	X2/Xn
O-FH	Uu

are unlocked from the proprietary network equipment. Virtualisation of network functions can take advantage of several methods. These are compared in Fig. 10, where four virtualisation options are illustrated together with the proprietary “black box”. Option number two from the left is to run the baseband processing software as “software on bare metal”, this opens the opportunity for the MNO to install the software on COTS hardware. However, no scalability improvements are foreseen since only one software system can run on one piece of hardware. The next opportunity is to utilise the scalability of virtual machines. This brings the opportunity of running a number of virtual machines on the same piece of hardware. This solution requires a hypervisor to be installed on the physical hardware to separate the physical resources into the required virtual system. Hypervisors can be of either type 1 or 2. Type 1 is installed on bare metal, whereas type 2 requires an operation system, and the method is referred to as hosted [66]. The last option is to install a container engine where network functions are installed as containers. Containers are very lightweight applications and contains only the functions necessary to run a specific network function. Hence, in comparison VMs are isolated systems but requires a lot of additional memory for various replicates where containers can install only the required functions on the selected machine and thus; has a higher utilisation of the physical resources.

Virtualised network functions are visioned to be able to run on any COTS hardware. This gives the MNO the flexibility to either install the COTS hardware on the cell site or centralise it in a data centre. In the latter case, the MNO can choose to run his own data centre or to rent capacity in a public cloud. Both solutions have benefits and drawbacks. The private data centre can implement a higher level of security where the public cloud solution is more flexible as resources can be rented on demand. The work in [67] discusses the maturity of the hardware to run virtualised network functions, as the authors claim that research has shown that vRAN requires many more resources than legacy RAN installations to achieve similar performance guarantees.

To keep up with the heavy processing demands of RAN functions in the physical layer, hardware acceleration is required to increase the efficiency while maintaining the flexibility from COTS hardware. However, dedicated accelerators make virtual baseband processing more expensive and power-hungry than their legacy counterparts [67]. The accelerator is used to offload and accelerate resource heavy computing tasks. Acceleration is used for vRAN to achieve optimal performance since the Central Processing Unit (CPU) can be reserved for the more complex operations. Hence, within virtualised RAN forums including O-RAN, there is currently a lot of attention to how CPUs handle physical layer processing [68]. Within vRAN two approaches are currently being discussed:

- Lookaside hardware acceleration
- Inline hardware acceleration

Some vendors, including Nokia, believe in inline acceleration, where other vendors, including Ericsson, believe in lookaside acceleration. The two approaches are described below and compared in Table 2.

Lookaside hardware acceleration describes hardware units that are integrated outside of the data path [71]. Hence, the lookaside accelerator is not an integrated part of the data path. In lookaside, the

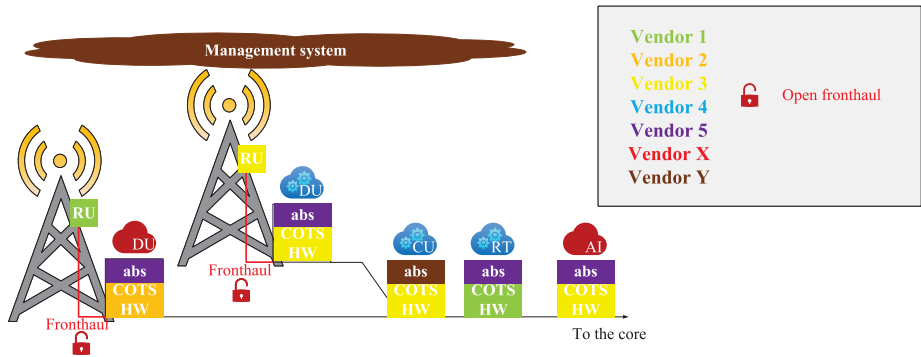


Fig. 8. Implementation of Open-Radio Access Network (O-RAN), utilising virtualised network functions, requiring an abstraction layer (ABS) between the hardware (HW) and Distributed Unit (DU) or Centralised Unit (CU) functions. The open lock refers to open fronthaul and the system comes with Real Time (RT) control and native Artificial Intelligence (AI) optimisation.

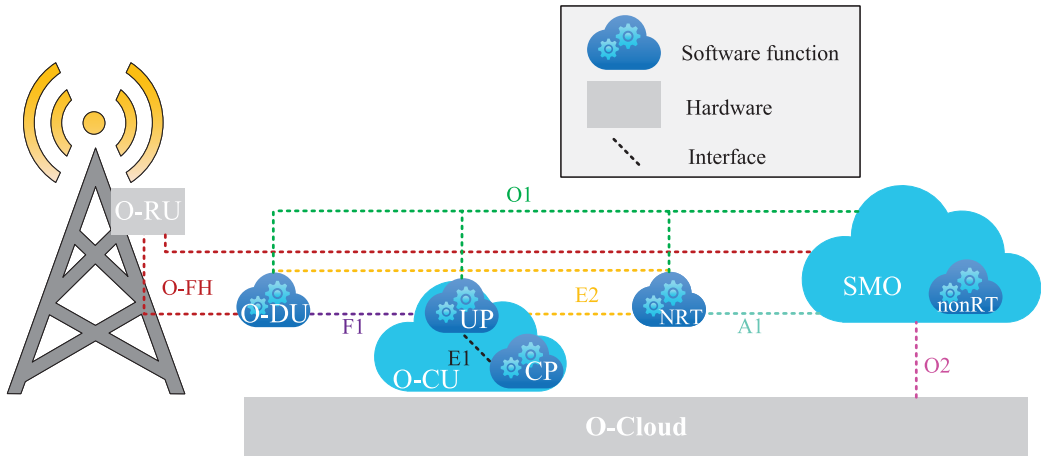


Fig. 9. The O-RAN logical architecture illustrating the various interfaces described in the O-RAN specifications and the components: Open Radio Unit (O-RU), Open Distributed Unit (O-DU), Open Centralised Unit (O-CU), Near Real-Time (NRT) Radio Interface Controller (RIC), non Real-Time (nonRT) RIC and Service Management and Orchestration (SMO) framework.

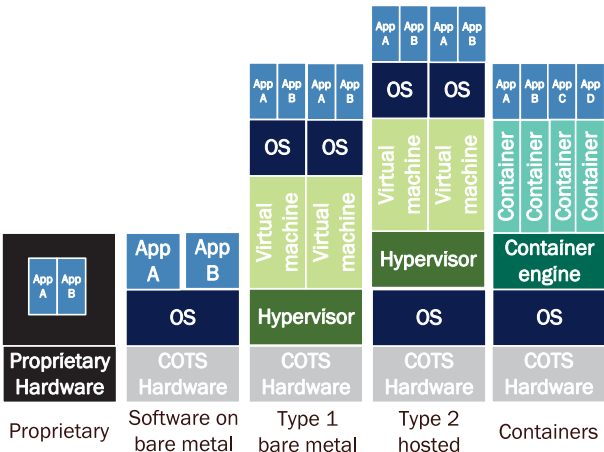


Fig. 10. Various options for baseband processing installation, from the left: Proprietary “black box” installation, bare metal installation, virtual machine installation with type 1 hypervisor, virtual machine installation with type 2 hypervisor and container installation. The figure illustrates the corresponding location of the Operating System (OS), and thus how much the hardware can be utilised.

accelerator is used for offload and only selected functions are sent to the accelerator. Thus, the lookaside accelerator does not have to work on the same throughput as the system. The selected functions are then processed by the accelerator and sent back to the CPU when

Table 2
Comparison of the two categories of hardware accelerators.

	Lookaside	Inline
Supporters	Ericsson, Intel [68]	Nokia [68], Azure [69], Huawei [69], Fujitsu [68], Marvel [70], Qualcomm [70]
Accelerated data	Selected parts of the physical layer	All physical layer
CPU tasks	L1+L2+L3	L2+L3
Benefits	- Multiple operations simultaneously [71] - mMIMO benefits [68] - High flexibility [70] - QoS support [72]	- Traffic pre-processed before CPU [73] - Separated accelerator independently scalable [73] - High capacity [73]
Drawbacks	- Physical layer processes consume CPU resources [73] - Lower capacity [73] - Overhead between accelerator and CPU [71] - Transport latency [72] - High Cost [68] - Complexity [72] - More CPUs	- All L1 accelerated before moving to higher layers [71] - Less efficient [71]

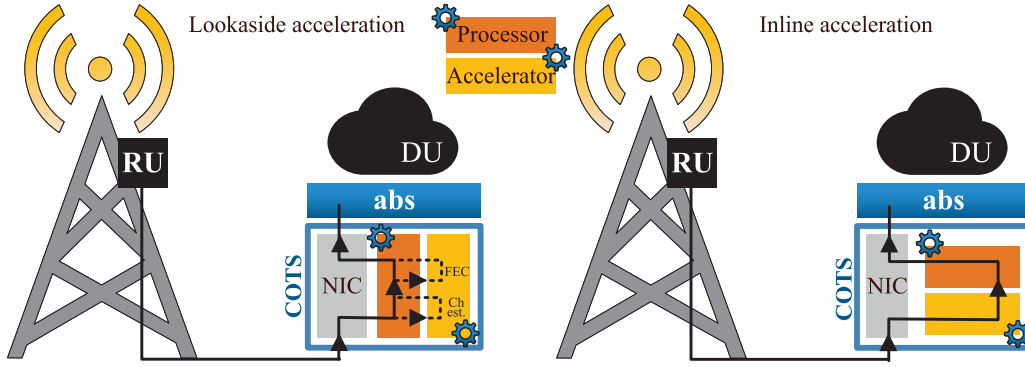


Fig. 11. A comparison of lookaside (left) and inline (right) acceleration. The figure illustrates the different dataflows for the two accelerator types, where lookaside only uses the accelerator for selected functions, channel estimation (ch. est.) and Forward Error Correction (FEC), then inline uses the accelerator for the whole L1 processing.

finished. These selected functions refer to the processes of Forward Error Correction (FEC) and Discrete Fourier Transformation (DFT), which are used in massive MIMO scenarios [68]. Other sources report the selected functions as the FEC and the channel estimation (ch. est.) [72]. The major benefit of lookaside acceleration is that multiple operations are handled simultaneously whereas a major drawback is the additional time spent on transfer between CPU and accelerator.

Inline hardware acceleration is tightly coupled with the rest of the system as specialised hardware components integrated directly into the system data path [71]. All the heavy physical layer processing is handled by an accelerator card. Hence, incoming data will continue to the CPU after L1 (physical layer) has been processed by the accelerator [71]. This leads to less data transmission between the CPU and accelerator card, due to the traffic being pre-processed by the accelerator before arriving in the CPU. This is both a benefit and a drawback since the CPU is reserved for higher layer processing while only one operation is handled at the time. The two accelerator opportunities are illustrated in Fig. 11, that shows how only selected functions are handled in the accelerator for lookaside whereas for inline the whole dataflow goes through the accelerator before continuing to the CPU.

Recent efforts in the area of accelerators focus on lookaside acceleration, where the overhead between accelerator and CPU is examined in [71], revealing how the length of the data sent to the accelerator significantly impact the performance of the solution. The work in [74] also report on the performance of the lookaside acceleration, here higher throughput is observed for smaller packets but also higher latency. A recent development in lookaside acceleration is Intel started to integrate the accelerator on the same chip as the CPU, referred to as an integrated accelerator [68]. This processor promises to combine inline benefits with lookaside benefits [70].

Fig. 12 illustrates the differences between processing using inline and lookaside acceleration. The figure shows how processing of a L1 message is handled by the accelerator for inline acceleration, whereas for lookaside acceleration only selected L1 functions are handled by the accelerator and the CPU is able to process the remainder of the message simultaneously. But also how lookaside has more transfers on the interface between CPU and accelerator. Furthermore, the figure illustrates how the use of accelerator improves the processing latency.

3.2.2. Centralisation

O-RAN supports both distributed and centralised processing. Both solutions have their own pros and cons, and adapt to different deployment scenarios and use cases [75]. Distributed processing functions, i.e. keeping all processing on the cell site, will benefit from lowered latency and lower requirements to the network connecting the cell site to the core. Centralising processing functions, i.e. as per the C-RAN architecture, can improve the cell cooperation speed as well as utilisation of hardware resources [62]. Hence, having many functions local as per the distributed architecture will put lower requirements to the transport

network in terms of latency and capacity, whereas centralising many functions will make better use of scarce resources being both hardware but also spectrum due to the increased cell cooperation. Centralisation can be implemented to different degrees, since the RU, DU and CU can be installed independently of each other, referred to as functional placement scenarios. The four different functional placement scenarios, distributed and centralised, supported by O-RAN and ITU-T [76] are presented in Fig. 13. The figure also illustrates the pros and cons of each option in terms of requirements to the transport network latency and capacity as well as the difficulties in finding a suitable DC location and the level of how much the resource utilisation gets improved. This is illustrated using bars where the colour green is pros and the colour red is cons. According to Fig. 13, then option 1 is most suitable in situations where the transport network is limited in terms of latency and capacity and there is no suitable locations for a Data Center (DC), but the drawback is that there are no centralised functions, and thus; no improved resource utilisation. Option 2 has a little more requirements to the transport network, but on the other hand more shared resources, since CU functions are moved to a DC. In option 3 and 4 the transport network requirements are very strict, but the resource utilisation is much improved. However, in option 4 two DC locations are necessary and it gets more complicated to find the right location. O-RAN only support options 1 and 3 for 4G [76].

As per the functional placement scenarios, the baseband processing can be centralised, or parts of it can be centralised. The decision is determined upon many parameters, but if the MNO is aiming for C-RAN centralisation benefits like improved resource utilisation and cell cooperation as well as a low site footprint, then it is crucial to look into the C-RAN prerequisites:

- Transport network capacity
- Central location for DC
- Latency compliance

In [77] O-RAN identifies the delay as having the largest impact on the network design, which is also one of the main drivers for using distributed processing. Thus, O-RAN proposes in [77] to handle CU UP processing for different types of services in different distances from the cell site. Different services being slices of Ultra-Reliable and Low Latency Communications (URLLC), Enhanced Mobile Broadband (eMBB) and Massive Machine-Type Communications (mMTC) traffic. Hence, all DU traffic is handled in a DC, located up to 20 km from the RU in an edge cloud location. The URLLC traffic then has its CU UP traffic executed together with the DU, as in functional placement option 1 or 3 in Fig. 13 (depending on distance) [77]. CU UP traffic for the eMBB can then be executed in a regional cloud up to 200 km from the edge cloud, as in option 2 or 4 in Fig. 13 (depending on distance between RU and DU) [77]. For the mMTC the CU UP traffic can be executed in a central cloud, up to 2000 km from the regional cloud [77].

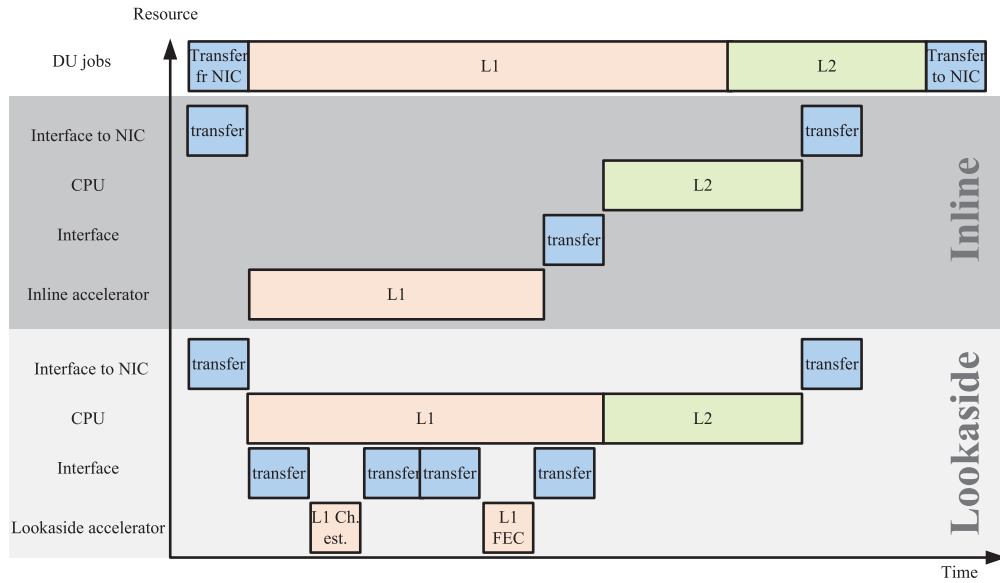


Fig. 12. A comparison of lookaside (bottom) and inline (top) acceleration. The figure illustrates how processing of the Distributed Unit (DU) handling L1 and L2 functions are carried out in each type of accelerator. The figure shows how all L1 processing is handled in the accelerator of type inline whereas only selected L1 functions are processed in the accelerator of type lookaside, requiring more transport effort on the interface between accelerator and CPU.

3.2.3. Open fronthaul

The open fronthaul, as stated in the O-RAN specifications, differs from today's CPRI fronthaul by having more standardised data planes. Thus, where the traditional CPRI fronthaul only has the UP and CP, then the open fronthaul has both UP, CP, a synchronisation plane and a management plane [76]. The latter handling non real time management flows over the open fronthaul, where the synchronisation plane ensures time synchronisation between elements of potential different vendors.

As described under open interfaces, the fronthaul faces a trade-off between RU complexity and fronthaul requirements. In the case of the FSs proposed by O-RAN Alliance, 7-2x cat A and B they benefit from a relatively simple RU, and thus; have higher requirements to the fronthaul compared to other FSs. The benefits and drawbacks for cat A and B in Downlink (DL) are highlighted in Table 3. Cat A is intended for rural areas with an upper limit of eight antennas, whereas cat B was intentionally designed to support massive Multiple-Input Multiple-Output (MIMO) scenarios in urban areas with up to 64 antennas [78]. The work in [6] raises a flag that implementing the precoding in O-RU will be at the cost of high complexity in the case of a high precoding stream number, leading to inter-symbol interference in O-RU caused by interference with non-desired users.

The solutions presented by FS 7-2x cat A and B occupy a fixed amount of capacity on the fronthaul network. This provides a low utilisation of the link, but also major vendors and operators reported disappointing results in Uplink (UL) testing for cat B [78]. The low utilisation of the fronthaul link is worsened since in FS 7-2x a number of streams equal to the number of antennas must be transmitted between the DU and RU [78]. This is because the processes channel estimation and equalisation are handled in the DU. The many streams to be transmitted over the fronthaul link require huge fibre resources available by the MNO. The critical UL performance was handled in the O-RAN alliance as a work item referred to as the UL performance improvement, which centred around two proposals: class A and B [78]. Class A includes the equaliser, channel estimation and beamforming wave calculation in the RU. On the other hand, class B proposes to keep the equaliser in the DU, while the channel estimation and beamforming wave calculation is handled in the RU [78]. Hence, the discussion points at whether the equaliser, balancing the channel interference, should be placed in the RU or DU. The argument leading to the class B proposal is to be able to gather signals from multiple sites and do equalisation across them. However, to avoid fronthaul overload,

Table 3

Comparison of the two categories of functional split 7-2x downlink.

Category	A	B
Purpose	Rural	Urban/MIMO
Benefits	- Simple on-site installation [79]	- Lowered fronthaul load [79] - Support for modulation compression [79]
Drawbacks	- Constant fronthaul load [62] - Higher fronthaul load [62] - Strict fronthaul latency requirements [62]	- More complex RU [79] - Constant fronthaul load [62] - Strict fronthaul latency requirements [62] - Inter-RU interference in O-RU [6]

channel estimation must be done twice, both in the RU and DU. FS 7-2x for UL with categories and classes are summarised in Table 4. The current decision in O-RAN alliance is that both classes co-exists as operation modes. Hence, RUs can be of either class A or B whereas the DUs must support both modes [78]. Fig. 14 illustrates the location of functions in the various categories and classes of FS 7-2x. In regard to the decision made, to support both operation modes, then all functions illustrated for the UL transmission in DU remain. Hence, for class B the channel estimation is done twice.

As illustrated in Fig. 14, the RU can be either cat A or cat B. But if the RU is cat B, it must support operation mode A or B. This results in three different types of RUs, which makes it complex for the MNOs. Both to understand the difference of the options but also to navigate in the outcome of selecting either one of them for a specific scenario. Hence, class A RUs are much more complex compared to class B RUs, which makes it harder for newcomers to enter the market, which has the risk of being dominated by few experienced vendors. On the other hand, the reason for selecting class A is the neat separation of functions proved better performance [78]. Thus, cat A RUs are the obvious choice for rural installation when following the O-RAN specifications and cat B seems to be a trade-off between smart functions and better resource utilisation. Class A and B are neither specified by O-RAN Alliance in [63,76,79,80] nor [81]. Hence, it is considered to be a matter that is still under internal alignment.

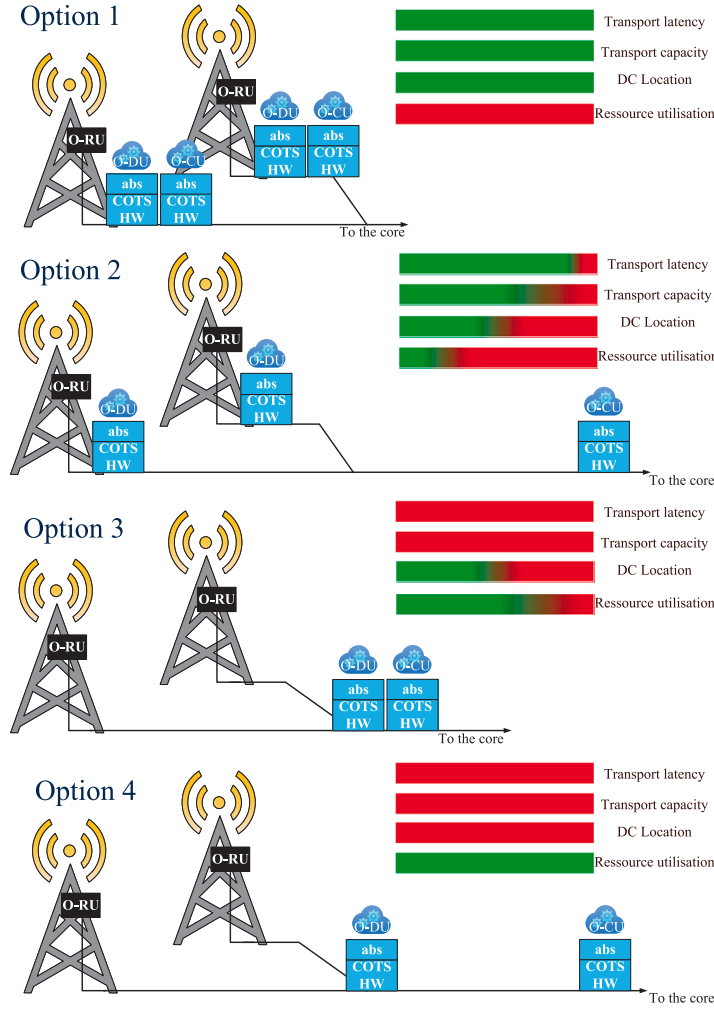


Fig. 13. Four O-RAN and ITU-T options for functional placement scenarios [76] illustrated with their respective pros and cons in terms of transport network latency and capacity requirements, the complications within selecting a good Data Center (DC) location and the opportunities for improved resource utilisation. In the bars red is a con and green is a pro, illustrating the various benefits and drawbacks of the different options. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 4
Comparison of the various categories and classes of functional split 7-2x uplink.

Category	A	B
Operation mode	Class A	Class B
Purpose	Rural	Urban/MIMO
Pros	- Simple on-site installation [79]	- High mobility [78] - High performance [78] - Neat cut-off [78]
Cons	- Fronthaul link requirements [79]	- Complex RU [78] - Expensive RU [78]

3.2.4. Native AI

One of the visions for the O-RAN ecosystem is to be AI-native. Hence, the foundation of the network is based on AI solutions, leading towards minimal human interaction. The processing-heavy AI duties are handled in the non-RT RIC together with the SMO framework

and applications are implemented as rApps. These rApps utilising AI functionalities are used for different optimisation options, below is an overview of some examples:

- Control management [82–88]
- Security [89]
- Performance management [90–92]
- Radio management [93–96]
- Energy consumption optimisation [97]

A part from above stated examples, the work in [98] shows the technical challenges in meeting QoS requirements for the RICs. Hence, to face these challenges in order to take advantage rApp optimisation in mobile networks, the MNO has to implement the non-RT RIC functionalities in a data centre large enough to handle the immense burden of AI workloads. The work in [99] utilises the simulation tool NS3 combined with ML to explore the capabilities of the O-RAN system. The complex network management process in O-RAN can be eased by an automation tool. Such a tool could be implemented as a rApp in the non RT RIC, like the “OrchestRAN” framework presented in [47]. This framework can be customised to match a specific MNO’s requirements in terms of for example network slicing, beamforming or scheduling control [47].

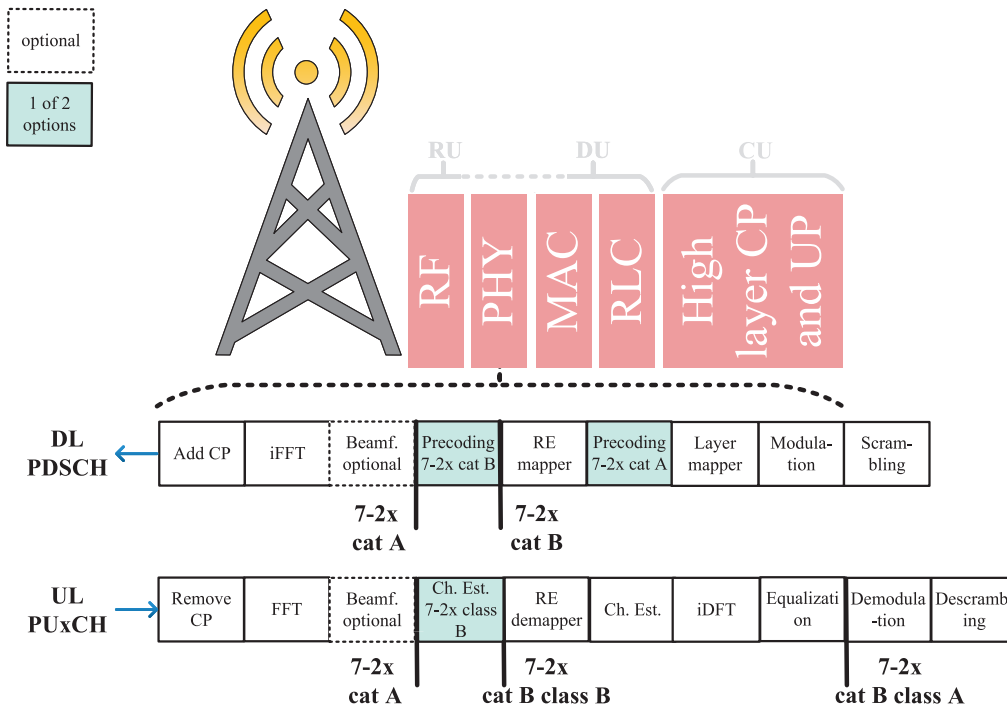


Fig. 14. An illustration of the functions in Radio Unit (RU) and Distributed Unit (DU) for the Physical Downlink Shared Channel (PDSCH) in Downlink (DL) and the Physical Uplink Shared/Control Channel (PUxCH) in Uplink (UL). The categories and classes of the UL split are shown to illustrate the functional division. The optional use of beamforming is illustrated by dashed lines and the two options for where to install precoding in DL and channel estimation (ch. est.) in UL are highlighted by colour. The figure is based on Fig. 4.2.3.1 1, Fig. 4.2.2.1 3 and Fig. 4.2.2.1 4 from [79], and Fig. 7-1, Fig. 7-2 from [80]. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

3.2.5. Intelligence

One of the ground pillars of the O-RAN ecosystem is to bring intelligence into the RAN. The RICs are key players in this vision, utilising AI and traffic data to make the RAN intelligent. The near RT RIC is placed in a control loop distance of <1s latency for the user [63]. On the other hand, the non-RT RIC is located further away, together with the SMO [63]. The non RT RIC is described as: "A logical function within SMO that drives the content carried across the A1 interface. It is comprised of the Non-RT RIC Framework and the Non-RT RIC Applications [100]." Hence, the non-RT RIC handles less timecritical processes such as AI/ML training, rApp management, OAM-related services and data management [101]. The near RT RIC is described as: "A logical function that enables near-real-time control and optimisation of RAN elements and resources via fine-grained data collection and actions over E2 interface [100]." The near RT RIC handles more time critical processes such as xApp management, tracking of UEs, listing UE associated data and end-point messaging [100]. Hence, the intelligence in the O-RAN system comes from the RICs making decisions based on AI/ML models and traffic data received by the near RT RIC over the messaging infrastructure. The near RT RIC also supports resolution of potential conflicts or overlaps of controls from xApps [100]. This provides a system that takes data-based decisions but also a system that can prioritise its decisions.

3.2.6. Security

By introducing O-RAN, many new tasks become the responsibility of the MNO, one of them being security. The introduction of O-RAN opens up for an expanded threat surface in the RAN with many new types of security attacks, for instance in the areas of virtualisation, native intelligence, and open interfaces [102]. In the latter case, a security threat could be a man-in-the-middle inserting a new clock to the front-haul synchronisation plane, but threats exist for all data planes and an exhaustive overview can be found in [103]. Examples of virtualisation security threats include virtual network function manipulation, virtual

machine mis-configuration or log leak attacks [102]. Examples of attacks related to native intelligence count denial of service, malicious data injection or manipulated AI training [102]. A comprehensive study of various security threats in O-RAN specific modules can be found in [102].

O-RAN creates and fills RAN specification gaps that expand the 3GPP standard. For example, the wireless Uu interface is standardised by 3GPP [104] and thus O-RAN follows the 3GPP standard in this interface for security including authentication, encryption, verification and so on [64]. The O-RAN specifications specify new components and interfaces not standardised by the 3GPP. These components are defined as "blocks" where security is an add-on. In [64] protection levels of critical assets are starting to be specified as normative security requirements with confidentiality, integrity, replay protection, authentication, and authorisation. However, the work in [105] proves how the RIC is still prone to attacks, by misleading it into triggering a malicious bearer migration procedure.

Nevertheless, as recognised by the EU Agency for Cybersecurity (ENISA) [106], standardised open interfaces could make it easier for auditors and security testers to understand how a certain RAN implementation is working. Hence, benefits of O-RAN security includes full visibility, the opportunity for the MNO to select the best modules, vendor diversity that decreases the risk of common coding errors or practices of single entities but also for the MNO to demand strong security capabilities from the supplier [102]. Furthermore, due to the software operations, the MNO can utilise a Continuous Integration and Continuous Deployment (CI/CD) operating model to fast fix detected vulnerabilities and network automation along with the use of AI can reduce the risk in human interaction [102]. ENISA concludes their study [106] with the recommendation of a cautious approach towards open RAN, leading to MNOs and other stakeholders should pay sufficient attention to ensuring security, which may require significant investments, on top of existing security measures.

4. Vendor management

In this section the expectations from a MNO towards an open multi-vendor RAN deployment are considered, together with the pitfalls a MNO might have to face when selecting partners for the open RAN journey.

4.1. Expectations

From a MNO point of view, open RAN is interesting in the perspective that it promises a higher degree of leverage towards the vendors. Hence, the MNO can mix and match equipment from various vendors or stick with one if that is the preferred solution. This is a promise of being able to select the equipment with a specific feature or a better price. However, MNOs operate within many models, some manage their own network and some use a managed service model. Hence, in the latter case the MNO will need new employees with a different set of skills to become a system integrator who manages his own network. On the other hand, the solution might not be to install open RAN in all of the mobile network. The installation can be limited to pico cells only, specific areas of macro and/or pico cells or to use open RAN for private networks. In all of these cases of limited installation, the size of the task is smaller, but in any case, a management system that can handle both the proprietary installations and the open RAN installations is preferred to avoid islands of various systems. Furthermore, MNOs often want to keep the equipment they already have installed and only change equipment that is broken or out of capacity. In this regard, opening the CPRI protocol could lead to the MNO only need to exchange the broken item and not all of the equipment installed on a site. However, this can also be overcome by a gateway.

4.2. Selection criteria

Implementing open RAN or O-RAN leaves many decisions to the MNO. This section states a number of criteria for the MNO to narrow down the process of selecting the right vendor or equipment. These criteria can be used as metrics in the process of evaluating potential vendors or products. Hence, the criteria can be used in any order, since the MNO might select a product after a specific feature or start off with a partnership before selecting products. Hence, the equipment or the vendor can be selected first.

4.2.1. Vendor criteria

Vendor management is a complex area and thus, this section provides an overview of selection criteria to be used for narrowing down the vendor landscape to open RAN partnership prospects. The criteria comes in different areas: Support, sustainability, economy and innovation. Hence, below list proposes a set of metrics to evaluate a potential vendor.

- *Innovation*; if the vendor delivers products with special features or newest technologies in the market.
- *Economy*; if the vendor is someone from who the MNO expects to buy multiple different products, the vendors general price range must be acceptable. Furthermore, the cost of transporting the products from the vendor to the MNO must be considered as well as potential tax and other regulations. Economy can also be impacted by the warranty the vendor provides.
- *Sustainability*; if the vendor live up to the MNOs Environmental, Social, and Governance (ESG) requirements for a vendor, and the vendor's subcontractors does too. Furthermore, the life cycle of products can be an important parameter to consider. A sustainability parameter may also be the transport distance for products and the materials that are used.
- *Support*; if the vendor has an easy-accessible support system.

4.2.2. Equipment criteria

When evaluating open RAN based equipment, the MNO must take at least three parameters into account: The price, the performance and the power consumption. These parameters must be equal to, or better than the current installation before open RAN is an economically feasible solution for a minor MNO. However, other factors do also come into play, an overview is found below:

- *The price* can be used as leverage or the MNO can choose the best one.
- *The performance* is important to consider, because a brownfield MNO does not want to reduce the performance of his network.
- *The power consumption* is also an important parameter, because even if a product has the same price and performance as an existing one, no MNO want more power hungry equipment.
- *Interoperability* is vital in an open RAN environment. The equipment's ability to inter-operate can be estimated if it has received an O-RAN certification or badge.
- *Sustainability* is measured in the material the product is made of or the expected life time, as well as the power consumption and energy efficiency.

5. Implementation strategy

The process of building an open RAN based network can be long-lasting. Thus, it requires an extensive roadmap to determine which order the deployment must follow. Concurrent with the MNOs and their vendors work, a lot of research projects are going on, expanding the knowledge in the field. An overview of these research projects can be seen in Fig. 4, which illustrates how a great deal of research projects were initiated in 2023 and thus, it will be very interesting to follow their results. Furthermore, many testbeds are instantiated for example by the projects: 6G-BRICKS, FIDAL, ELIXIRION and 6G-XR where open RAN based equipment will be explored long before roll out. On the MNO side, before the deployment can initiate, there is a number of things that must be examined, initially by integration tests and later in a sandbox test environment. Hence, for the MNO the goal is to gain practical experience with live network user traffic without affecting too many users, in case anything goes wrong. The implementation strategy initiates three phases which will be described in below subsections, to initiate an open RAN deployment roadmap for a brownfield MNO.

1. Sandbox testing
2. Prior practicalities
3. Initial deployment
4. Large scale deployment

5.1. Sandbox testing

A starting point is to examine what degree of openness is expected, since a vRAN solution can also be used by the MNO as a stepping stone to gain familiarity with operating a virtual network and manage more than one vendor. The location most suitable for sandbox testing, in the live network, is a place where most users are not affected by the tests. Thus, an area where both coverage and capacity layers are installed can be considered, because if using a capacity layer for open RAN testing, users will still have coverage in case anything goes wrong. However, the traffic must be low in the current area for the coverage layers to be sufficient, and the site must be easily accessible since there is a potential for many visits by technicians during the test phase. Thus, requirements for sites to be candidates for open RAN sandbox testing is listed below:

- Site accessibility
- Coverage and capacity layers
- Low traffic

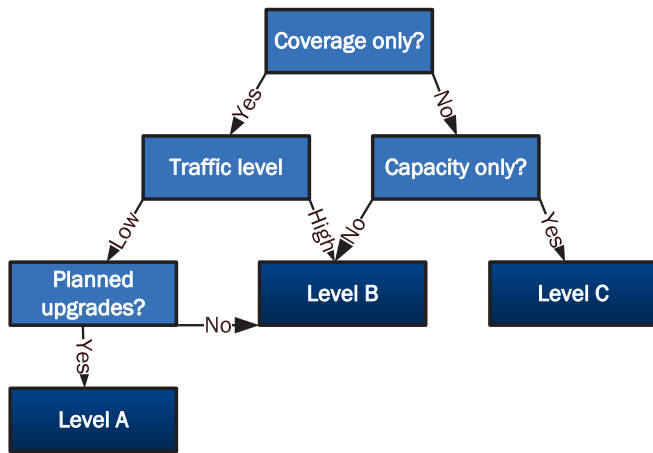


Fig. 15. Decision tree for dividing cells into three levels: A (No initial open RAN deployment), B (limited suitable for open RAN deployment) and C (suitable for initial open RAN deployment).

5.2. Prior practicalities

During the period of sandbox testing, the MNO can gain familiarity with running a virtualised network and then a multi-vendor network with selected partners. However, before installing open RAN equipment in the network, a number of factors must be in place:

- A complete end-to-end overview of the network, including products used and software versions
- A management system that can handle both open RAN and proprietary equipment
- An engineering department ready for becoming their own system integrator and fault inspector. i.e. new skills are required to handle new tasks
- A vendor management department ready to handle multiple vendors. i.e. Multiple vendors also bring complications in product updates, fault management and many portfolios to consider

The above list shows some of the concerns that a smaller MNO might consider when exploring the opportunities within open RAN, and thus; opens the opportunity for new business and research areas. Another aspect, that must be in place during this phase, in case the O-RAN specifications are followed, is the virtualisation. Virtualisation of the RAN adds very large requirements to the underlying hardware, which is the focus of the CORENext project (see Section 2), that works towards efficient, scalable and virtualisable accelerators. The factors pointed out above, are critical initiators of a multi-vendor environment, and especially the end-to-end overview of the network and the compliant management system, cannot be compromised.

5.3. Initial deployment

In order for a brownfield MNO to select the site position, where it brings most value to initialise an open RAN deployment, the area where open RAN brings most benefits must be identified. In this work, sites are categorised into level A, B, and C. Level C are the sites not suitable for initial open RAN installation, level B are the sites suitable for partial open RAN installation or later installations and level A recognises the cells suitable for initial open RAN installation. Furthermore, sites that are prospects to upgrades of the existing equipment should be considered first. The open RAN vendors, who have entered the market within the last years, are still to some degree inexperienced and thus; simpler installations will be the obvious choice. A simple installation covers passive antennas and small antenna arrays, as is often used to provide the coverage layer of deployment in rural areas. Hence, features to select cells for initial open RAN deployment are:

- Planned upgrades
- Coverage layer-only installation
- Low traffic

Furthermore, in case the MNO also wants C-RAN then the list should also prioritise high capacity fronthaul opportunities and the availability of sites suited to host a cloud datacenter within the required distance.

5.4. Large scale deployment

The cells in level B are those that will be upgraded later or has not only coverage layers but also capacity layers, hence; only some of the equipment can be exchanged with current open RAN level of maturity. The decision tree in Fig. 15 shows the selection process of finding the cells prospective for open RAN deployment. Building and maintaining a large scale open RAN network, requires new ways of operating the network. Large research projects will contribute to evolving the ecosystem (see Section 2). For instance the O-RANOS project will leverage a rApps and xApps development framework, which will evolve the landscape of more vendors and more diverse solutions. The YO-RAN project develops amongst other components a RIC, which is vital for the vision of the O-RAN specifications.

5.5. Case study

This case study considers the mobile network of TDC NET, a MNO with 99% 5G coverage in the country of Denmark. TDC NET operates more than 4000 sites, and owns licenses to many frequency bands, both those below 1 GHz used for coverage layer and those above 1 GHz used for capacity layers. This case study investigates sites that are prospective candidates to open RAN deployment. Thus, information about the various sites' installations, traffic and positions were collected from different systems and analysed to rank the various sites.

According to the decision tree provided in Fig. 15, the first step is to consider the coverage and capacity layers of the various sites and sort away those with capacity layers. The result of that process is illustrated in Fig. 16, in the upper right corner. The figure shows how 9% of the sites only have a coverage layer. These 9% of the sites are considered alone when moving forward in the ranking process.

After the initial selection process, the traffic load must be considered. Here the sum of 4G and 5G traffic for all cells on one site were considered in a time range of one day. Then the 25 sites with the lowest traffic, out of the 9% coverage layer only sites, were selected. Fig. 16, shows the 25 selected, coverage only, low traffic sites, which are spread all over the country of Denmark. Thus, more than 4000 sites were narrowed down to 25 candidate sites, prospects for open RAN deployment. Hence, if TDC NET wanted to install open RAN equipment in their network they would consider which of these 25 sites would be up for upgrades of the equipment.

6. Discussion

What is open RAN? According to the definition stated earlier in this paper, the interfaces between network elements must be publicly declared, or even standardised. From a brownfield MNO perspective, then a full scale network is already deployed with CPRI and enhanced CPRI (eCPRI) RUs installed. Thus, the migration to an open RAN installation will be either carried out when network equipment shall be exchanged anyway or a full network swap is required. The latter is though not realistic with the current maturity of O-RAN. Hence, open RAN vendors cannot yet promise the same performance as the proprietary vendors who are much more experienced.

The roadmap towards open RAN implementation has multiple facets, which has been described throughout this paper. First of all, it requires the MNO to take an active decision regarding the network design in terms of component specifications. This process is supported in Section 3. Then the MNO must decide what deployment model to go with,

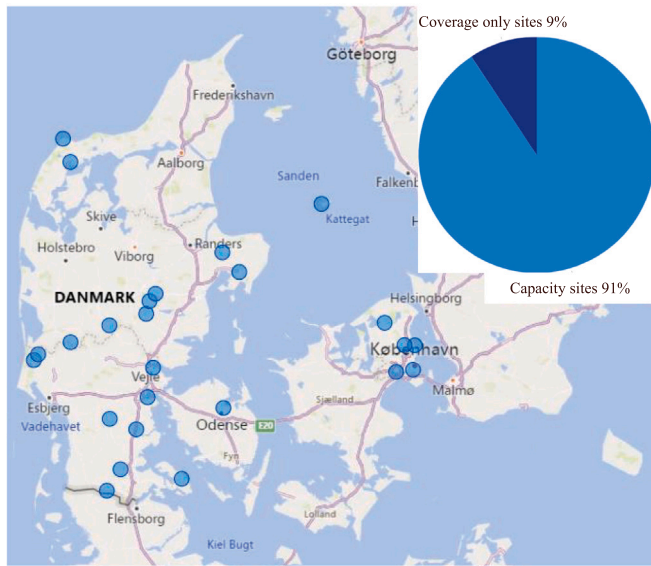


Fig. 16. Upper right corner shows the relation between sites with coverage only and sites with a capacity layer. The graph shows how 9% of the total amount of sites only have a coverage layer installed. The other part of the figure shows a map of Denmark, pointing out the locations of the 25 most suitable sites for open RAN deployment, out of 4100 total sites in the network. The map is generated using MS power BI.

including the degree of openness and how many vendors should be present in the network. This process is supported by Section 4. Finally the MNO must make a plan, derive a fulfilling roadmap towards the deployment, which includes actions to be taken prior to the deployment but also test phases. This process is supported by Section 5. Hence, this work provides a comprehensive overview of actions and decisions to be taken before initial open RAN implementation.

Considering the deployment phases, both the sandbox and initial deployment, possess very low risk for the MNO. A site with more traffic would possess higher risk, since more users could be affected if anything goes wrong. But will these very low risk scenarios result in less learning? The reflections have many faces, however; we believe that from the scenarios proposed there is much to learn. Hence, not just gaining familiarity with running a vRAN and the integration and deployment of multi-vendor equipment but also to gain experience with managing multiple vendors in the network, and thus; areas that are not only affecting the engineering department.

6.1. Open RAN MNO evaluation

To achieve open RAN, by means of open interfaces in the RAN, a publicly declared way of defining the interfaces must be accessible for all potential vendors. O-RAN provides a set of public and international specifications, which is a crucial step towards achieving open RAN. From a MNO's point of view, O-RAN has still many loose ends. But when breaking things down, the MNO is merely interested in one thing: To deliver the best Customer Experience (CEX) for the lowest cost. Thus, Fig. 17 explores the potential pains and gains from a MNO perspective when deploying O-RAN. The objectives are divided into three categories, determined after implementation effort, highlighted by colour where green represent minimum implementation effort, orange represents medium and red represents maximum. The graph illustrates how the objectives are expected to affect the CEX as well as how much cost savings they are envisioned to bring. Hence, in the case of C-RAN, it is expected to bring small improvements in CEX and cost savings, however; it requires much effort to centralise the processing and build the fronthaul network. Security is red, because it requires much effort from the MNO to secure the O-RAN based network, as the

MNO himself is responsible for securing all layers of the network. This is expected to cost more money than legacy networks and it might have a negative impact on the CEX, since security issues are a risk and solutions require much processing capacity [107]. COTS hardware possess a risk that it might degrade the CEX since the equipment is not specialised for the purpose of RAN. The equipment might be cheaper in cost-price but it will be quite expensive to make it run in the network. Including more vendors in the network, might improve the cost, but vendor diversity does not necessarily improve the CEX. In the opposite end of the scale, the use of AI and CI/CD operations is expected to improve both cost savings and CEX. Below subsections will describe the benefits and drawbacks O-RAN adoption will have for a MNO. The section is finalised with a set of open questions for O-RAN adoption.

6.1.1. MNO benefits

There are many reasons to the buzz around open RAN and O-RAN specifications. Hence, major benefits of O-RAN introduction include:

- *Vendor diversity* recognises the old saying “Do not put all your eggs in one basket”. This means that having multiple vendors by spreading the investments will reduce risk.
- *Native AI* is the futuristic vision of the O-RAN ecosystem. There is no doubt that AI is getting more and more used while its capabilities are continuously improving. Thus, being able to utilise AI leading towards zero touch networking, will minimise human faults and latency, this way it will improve both CEX and OPEX.
- *CI/CD opportunities* combine the practices for operations and development leading to frequent delivery and integration of new code. Hence, the software is continuously updated and improved.
- *Faster time to market* gives the MNO the advantage of being able to implement the newest ideas and improvements to the mobile network.
- *Potential CAPEX reductions* will be the result of a more diverse market. It will give the MNO the power to mix and match equipment from different vendors, which is an advantage in a price negotiation situation.

6.1.2. MNO drawbacks

Some parts of the industry are very eager to promote the advantages of open RAN and O-RAN, but the solutions also come with a backside. The white paper [58] defines four main challenges for brownfield operators, ie. operators with a legacy installed network, to adopt O-RAN: Integration of open interfaces and RIC; integration of O-RAN RU; dis-aggregation and virtualisation of network functions as well as hardware performance. The following list provides major drawbacks of O-RAN as well as the potential solutions:

- *Vendor management* is a tedious process, and when the amount of vendors is somehow unlimited, the MNO needs to establish his own selection criteria to narrow down the process, as described in Section 4.2. *The potential solution* is to hire more people to handle this process, or to use single-vendor O-RAN.
- *Product integration* requires a lot of testing and potentially also tailored to suit a specific deployment. Furthermore, when some components are updated, the updates must be integrated into the system and aligned with products from other vendors. *The potential solution* is to invest in equipment with O-RAN certifications or badges [42]. Additionally, buy an integrator service which manages multiple vendors and ensures fault management by proposing performance guarantees. The concept is illustrated in Fig. 18, showing how the integrator is responsible for integrating the open and proprietary parts of the network as well as configuring the system and manage the vendors. However, buying such a solution, it is not clear how it will be economically feasible to pay yet another vendor. Current examples of system integrators count Rakuten being system integrator for 1&1 in Germany [56], and Ericsson becoming system integrator for AT&T in the US [108].

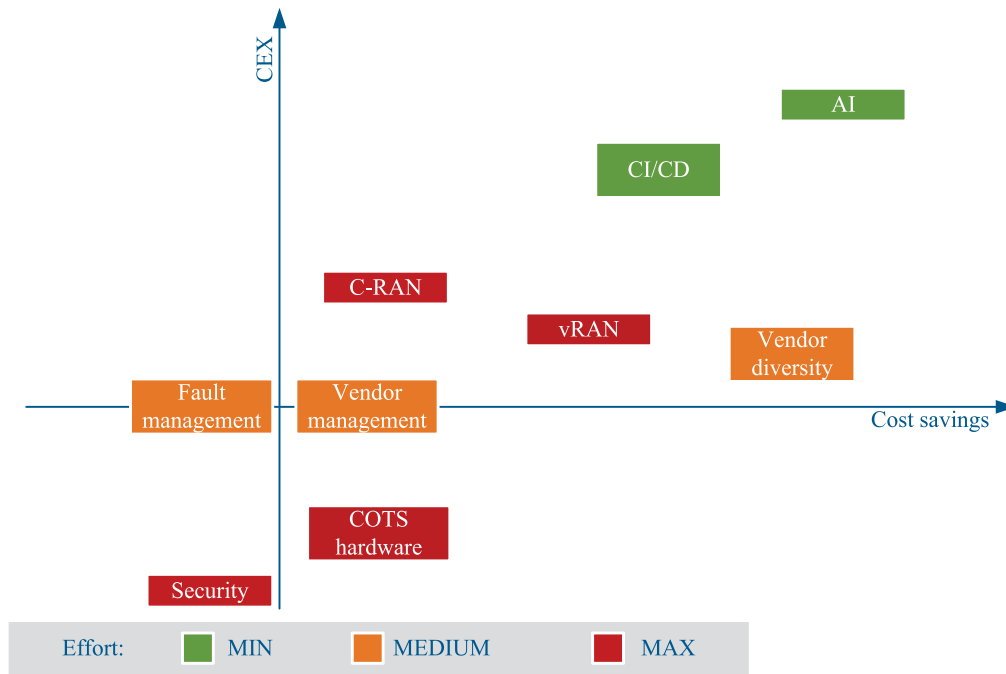


Fig. 17. The pains and gains for a Mobile Network Operator (MNO) if installing O-RAN, in the perspective of Customer Experience (CEX) and potential cost savings. The figure is colour coded according to implementation effort required. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

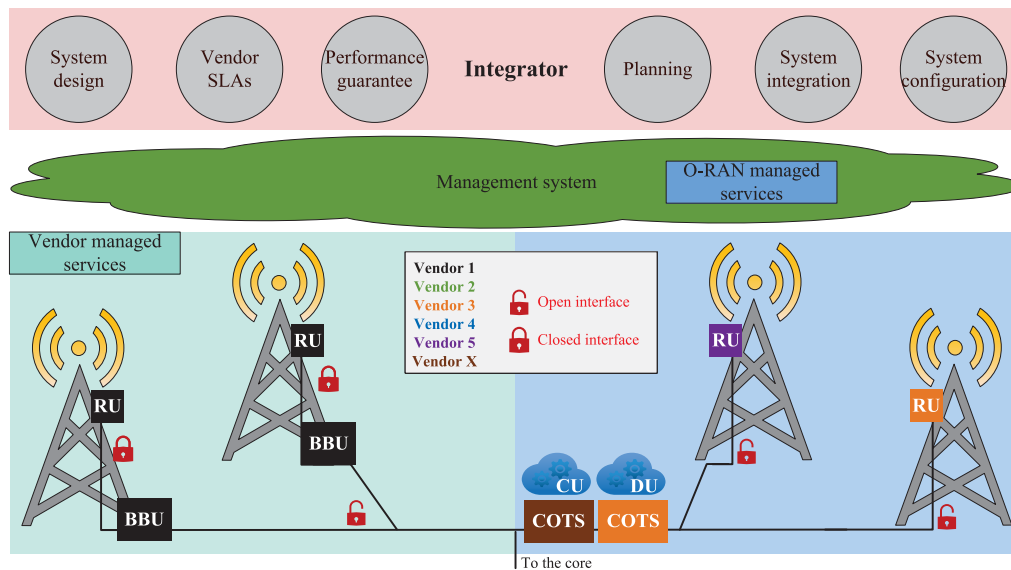


Fig. 18. O-RAN opens up for many new business opportunities, one of them is to become a system integrator for the MNO. The figure illustrates how the system integrator integrates and configures the system, and manages the multiple vendors for the MNO. The left part of the figure illustrates a proprietary system with closed interfaces between Radio Unit (RU) and Baseband Unit (BBU), whereas the right part of the figure illustrates a virtualised network with open interfaces.

- **Security** adds another layer of complexity to running an open RAN mobile network. However, in [109] mature security protocols like IPsec are required for implementation. IPsec is investigated for O-RAN implementation in [107] where authors highlight an O-RAN trade-off between security and processing power. *The potential solution* is that the MNO must be aware of this and employ experts in the field to secure the critical infrastructure. This is, however; a complex process since all layers must be properly secured. The O-RAN security workgroup specifies in [64] a set of security requirements for the network functions, applications and interfaces in the O-RAN ecosystem. This is though based on

more generic requirements of for instance authentication and encryption procedures, whereas it does not clearly propose specific implementations. The work in [103] proposes macsec to secure layer 2. The work in [110] uses the RIC for early detection of Denial-of-Service attacks. A solution can also be for integrators to offer and ensure this service.

- **Network management** is envisioned to be the next big issue for MNOs adopting open RAN [111]. The question is, how to make equipment, for instance RUs from different vendors inter-work? *The potential solution* is the SMO [111]. For a brownfield operator, the transition to O-RAN will not happen over night, and maybe it will only be installed in selected areas. Thus, requirements to

such a management system will be to support both O-RAN and proprietary types of installation, since they will have to co-exist. Rakuten offers their own system [112], but also NTT Docomo offers a SMO [111] for automated network maintenance. Another OAM system is designed in [113].

- *Fault management* and network management gets complicated when multiple vendors are delivering the network. Because, who is responsible for the error that occur, and who will fix it? *The potential solution* is to acquire a fault management system. The work in [114] utilises AI to provide a unified intelligent framework for multi-vendor and multi-domain fault management. A proposal to make the network more fault tolerant will be to use techniques such as state partitioning, partial replication, and fast re-route with role awareness to decrease the overhead for improving scalability and fault-tolerance in the RICs xAPPs [115].
- *COTS hardware* is widely discussed these days. It is still unknown if vRAN running on COTS hardware will perform like proprietary installations in run times, latency or energy consumption. Furthermore, the current discussion, as pointed out in Section 3.2.1, between using inline or lookaside accelerators is a crucial decision on the direction future COTS hardware for vRAN will take. A discussion on capacity versus flexibility. The major benefit on inline processing is its acceleration of all L1 processing, which might be a benefit in cases where much L1 processing is handled in the DU, such as for FS 7-2x cat A. On the other hand, the benefit of lookaside acceleration is the flexibility and benefits for mMIMO, which might be a benefit for FS 7-2x cat B. Hence, in the future there might be room for both directions, however; this requires very area specific deployments. *The potential solution* is complex, since the requirements to the hardware are linked to the choice of the FS. The more physical layer functions to be handled in the DU, the higher requirements to the DU hardware. The question is, should we build software that matches the capabilities of the hardware - or should we build hardware that matches the capabilities of the software?
- *Policy complications* is a sensitive area to step into. However, a lot of O-RAN development is arising from the fact that several governments rise security concerns against Chinese vendors [116]. Sweden for instance have banned Chinese firms from 5G networks [117]. This can potentially limit the openness of the open RAN networks, if not all vendors are considered. Furthermore, it is a lot of vendor management to ensure the whole chain of vendors are complying with country and company policies. Following the U.S., China has the second largest number of members in the O-RAN Alliance [116]. Many Chinese companies under U.S. sanctions are involved in the alliance and China Mobile, as a founding member, has permanent seats on the board of directors and veto rights [116]. Hence, China is deeply involved in the O-RAN alliance, and also with relations to the government since China Mobile is state-owned [116]. Thus, it is somehow contradictory that the rationale of O-RAN interest in many countries was to get out of the relation to one Chinese vendor, whereas the solution seems to be an alliance including many Chinese vendors. *The potential solution* is a political matter, out of the scope of this work.
- *Legacy equipment* is a matter all brownfield MNOs must take into account. Brownfield MNOs face other issues than the MNOs building a modern network from scratch. Hence, legacy equipment can infer a complication for MNOs exploring the opportunities within O-RAN specifications that only covers 4G and 5G Radio Access Technologies (RATs) [28]. In this regard legacy equipment can be 2G and/or 3G networks that are still crucial to the network operation. From this perspective, it might complicate network operations to move parts of the network (4G and 5G) to a multi-vendor system while still running a proprietary legacy installation. Furthermore, centralising components from 4G and

5G systems for a minimised site footprint will not have as much impact when legacy installations are left on the cell site. *The potential solution* could be to open up the CPRI protocol. Hence, to open the CPRI management plane. This would make current installations able to inter-work with equipment from other vendors, without having to swap the whole site installation. This solution will also make MNOs able to install gateways, of their own choice to support the FS they wish.

- *Visibility* of the O-RAN alliance is hard to get when you are outside the alliance. However, the cost of joining the alliance is the same for all sizes of MNOs and an annual membership fee of \$50,000.00 is a lot of money for smaller MNOs. This can leave some operators out of the alliance, where they will have less visibility of what is happening in the community. *The potential solution* will be to lower the membership fee for operators below a certain number of customers or to make an alliance for smaller MNOs where they can participate under a joint membership. Hence, a number of small MNOs would have the opportunity to split the membership fee amongst them, to get access to the alliance.
- *Complexity* of the network design is increased by the newest trends. Both the necessity for MNOs to choose between inline and lookaside acceleration as well as the various categories and classes of RUs makes it more complex for MNOs to design their networks. The more complex solutions, such as lookaside accelerators and cat B RUs makes it harder for new vendors to enter the market, as they naturally have less experience compared to companies who have developed RAN equipment for 20–30 years. However, it also makes it difficult to balance between openness and network performance. *The potential solution* could be for O-RAN to make a clear recommendation, but it could also be to include a larger degree of visibility in terms of what the different proposals bring. Hence, what are the benefits and drawbacks of the different solutions should be visible in the O-RAN specifications.

6.2. Outlook

Due to the large buzz around open RAN and the O-RAN alliance from both industry and academia, MNOs might be afraid of lacking behind if they do not explore open RAN opportunities. However, as pointed out in the case study, many things are yet preventing smaller MNOs from investing in the O-RAN technology. This being for instance lack of large scale deployment cases, mainly due to: Technology and products are not yet competitive, immature systems integration and operation models, the implementation timing needs to fit with RAN investment cycles [118]. Furthermore, being a small MNO, we are left with a number of open questions:

- How large does an operator need to be, to benefit from open RAN?
- What is the cost of COTS hardware in performance and power?
- How can a MNO balance the open RAN compromise between price and quality?
- How many employees would be needed for open RAN adoption, and what skills should they possess?

7. Conclusion

Embracing open RAN is not just a strategic choice; it is a transformation journey that demands carefully planned design and informed decision making. Hence, this paper provided a thorough examination of the critical steps a MNO must undertake to prepare for open RAN installation successfully. Thus, the foundation of this work is to prepare the MNO to navigate in increasingly complex design opportunities, multi-vendor strategies as well as road-mapping the deployment. This includes decisions upon centralisation or more functions at the edge

and knowledge about what impact the choice of FS has on the network as well as the choice of accelerator type. The vendors, who deliver the selected solutions, must not only be chosen because they have the best product but also based on their innovation level and sustainability. Finally, benefits and drawbacks of MNO O-RAN adoption were discussed, highlighting integration challenges, lack of visibility but also the increasing complexity. This complexity does not only make it more difficult for MNOs to navigate in the options but also for new vendors to enter the market. In conclusion, the successful adoption of open RAN technology represents an opportunity for MNOs to enhance network flexibility, reduce costs, and accelerate innovation. However, this transition demands careful planning and design of the network as it is built up of modules which individually must be diligently arranged. Hence, by attentive preparations, MNOs can embark on their open RAN journey with confidence.

CRedit authorship contribution statement

Line M.P. Larsen: Writing – review & editing, Writing – original draft, Visualization, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Henrik L. Christiansen:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization. **Sarah Ruepp:** Supervision, Funding acquisition, Conceptualization. **Michael S. Berger:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

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Data availability

The data that has been used is confidential.

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